Stress-strain-temperature relations

Chapter 5

Tensile test

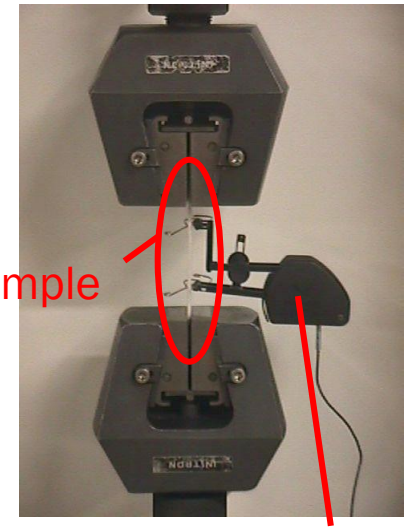
- One of the simplest tests for determining mechanical properties of a material is the tensile test. In this test, a load is applied along the longitudinal axis of a circular test specimen. The applied load and the resulting elongation of the member are measured.

The tensile test is an extremely important experimental technique, used to determine Young's Modulus, and other material properties.



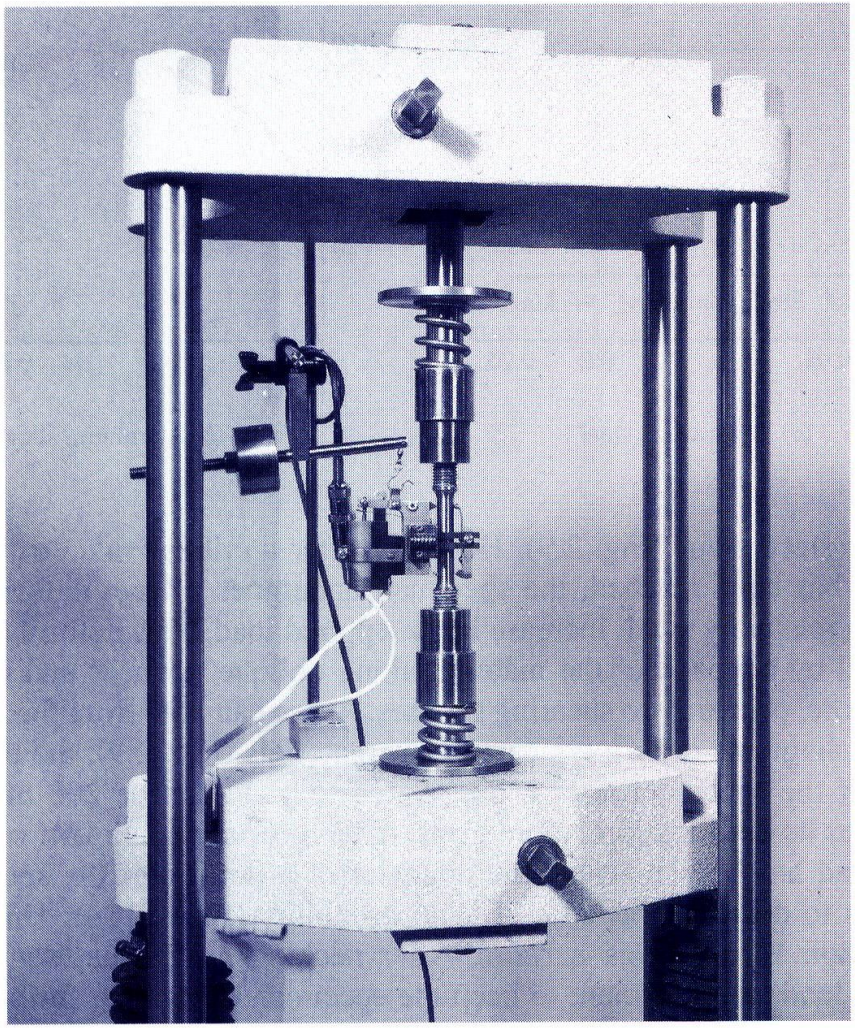


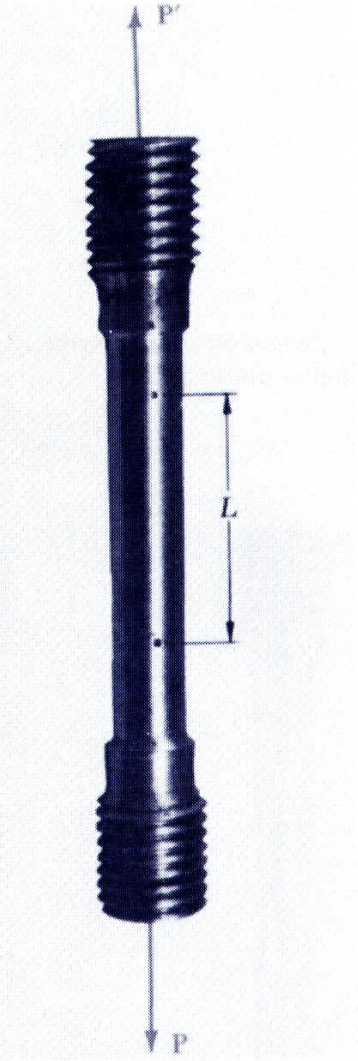
A sample is stretched at a constant rate.

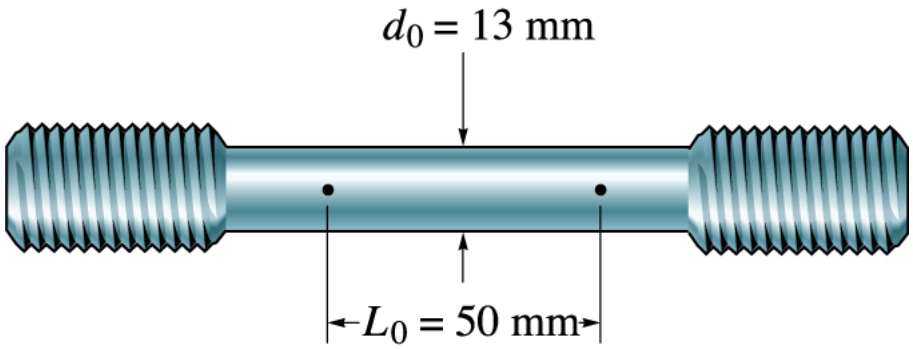


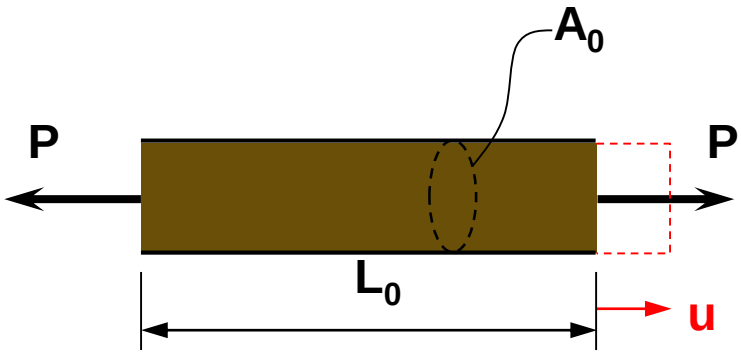
Required tensile force is measured, along with the changing dimensions of the sample.

Stress-Strain Test









Engineering stress and **engineering strain** are calculated as:

$$\sigma = \frac{\text{Internal Normal Force}}{\text{Original C.S. Area}} = \frac{P}{A_0} \quad \text{(Pa)}$$

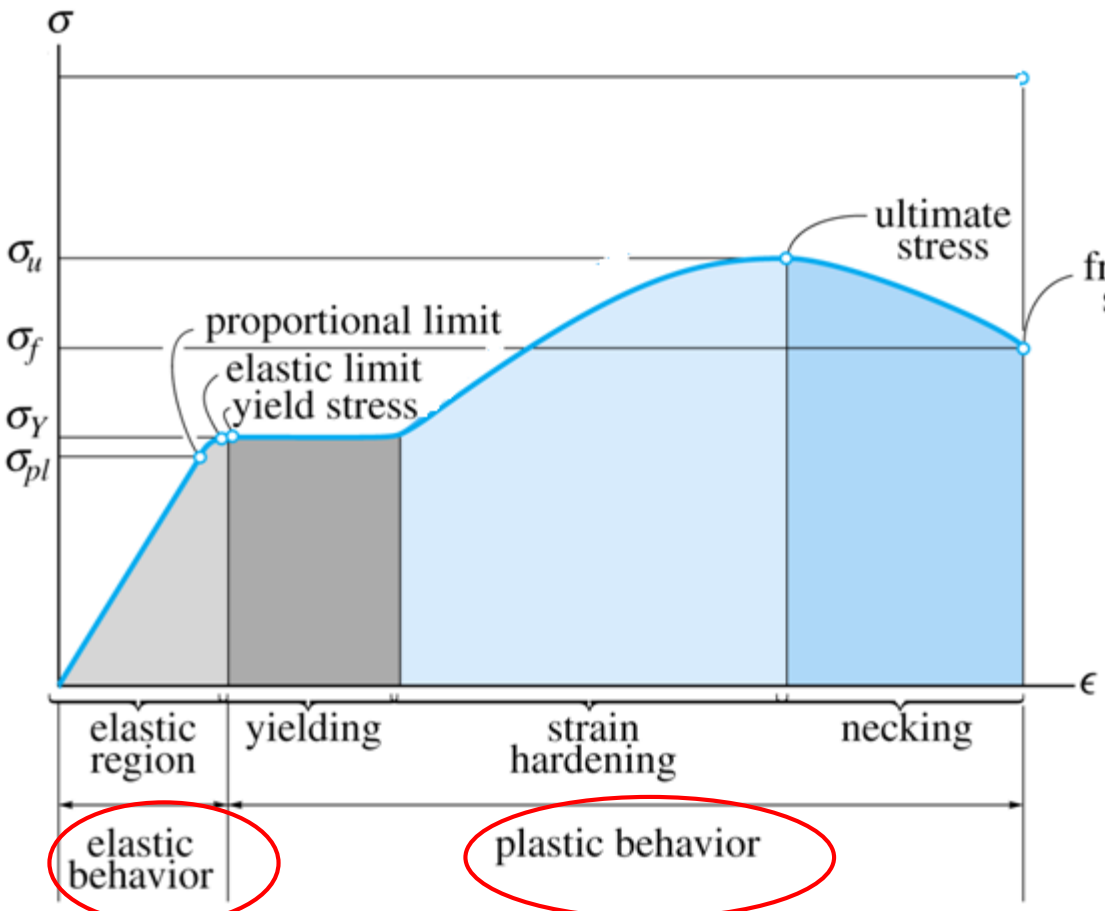
$$\varepsilon = \frac{\text{Change in Length}}{\text{Original Length}} = \frac{\delta}{L_0}$$

(Dimensionless)

These are the measures most commonly used in practice. However, the **true stress** and **true strain** are found by considering the current sample cross-sectional area and length.

$$\sigma = \frac{P}{A} \qquad \varepsilon = \frac{u}{L}$$

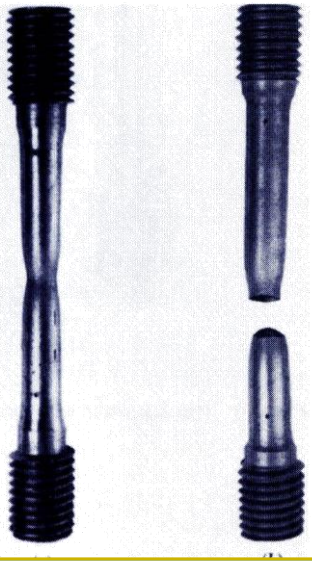
Stress-Strain Diagram: Ductile Materials



Conventional and true stress-strain diagrams for ductile material (steel) (not to scale)

Recoverable deformation

Permanent deformation



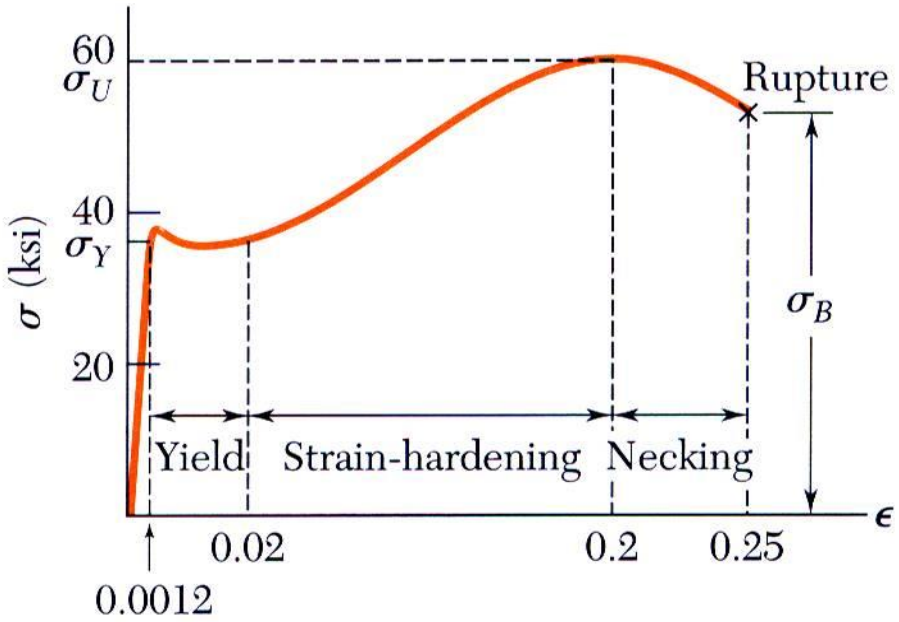
Materials that are capable of undergoing large strains (*at normal temperature*) before failure. An advantage of ductile materials is that visible distortions may occur if the loads before too large. Ductile materials are also capable of absorbing large amounts of energy prior to failure. Ductile materials include mild steel, aluminum and some of its alloys, copper, magnesium, nickel, brass, bronze and many others.

Mechanical properties of materials

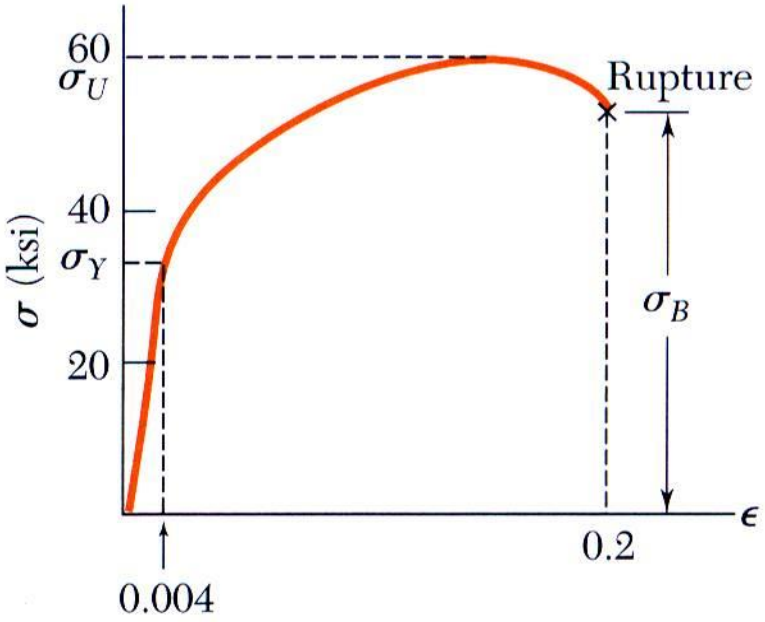
- **Proportional Limit** - Stress above which stress is not longer proportional to strain.
- **Elastic Limit** - The maximum stress that can be applied without resulting in permanent deformation when unloaded.
- **Yield Point** - Stress at which there are large increases in strain with little or no increase in stress.

Mechanical properties of materials

- **Fatigue -** A member may fail due to *fatigue* at stress levels significantly below the ultimate strength if subjected to many loading cycles.
- **Creep** – Property by which a material undergoes deformations at constant stress over time.
  - Time dependent part of deformations
  - A body undergoes elastic deformation immediately on the application of load. This deformation increases over period of time due to creep.

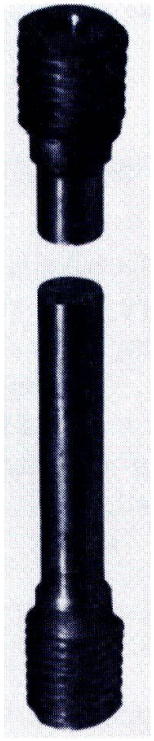


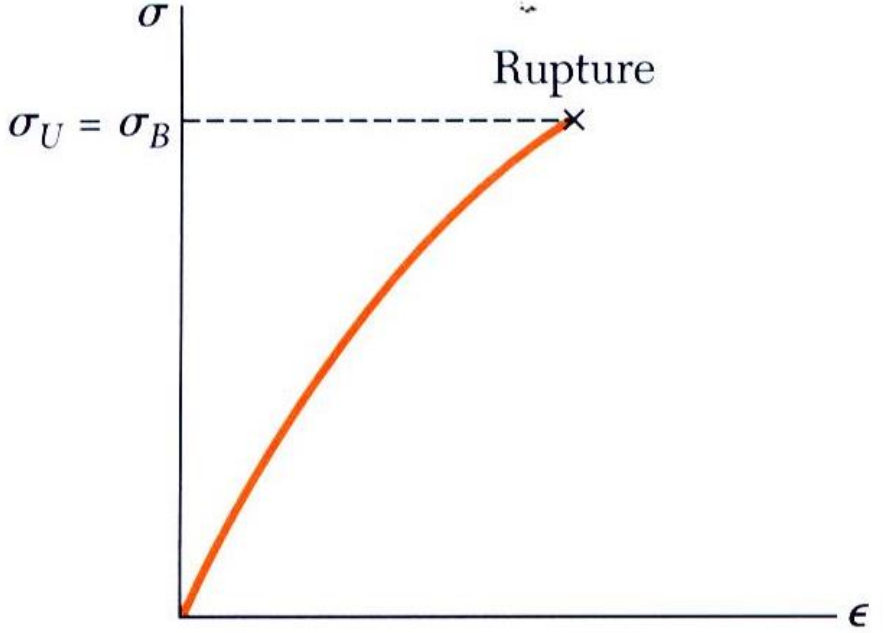
(*a*) Low-carbon steel



(*b*) Aluminum alloy

Stress-Strain Diagram: Brittle Materials

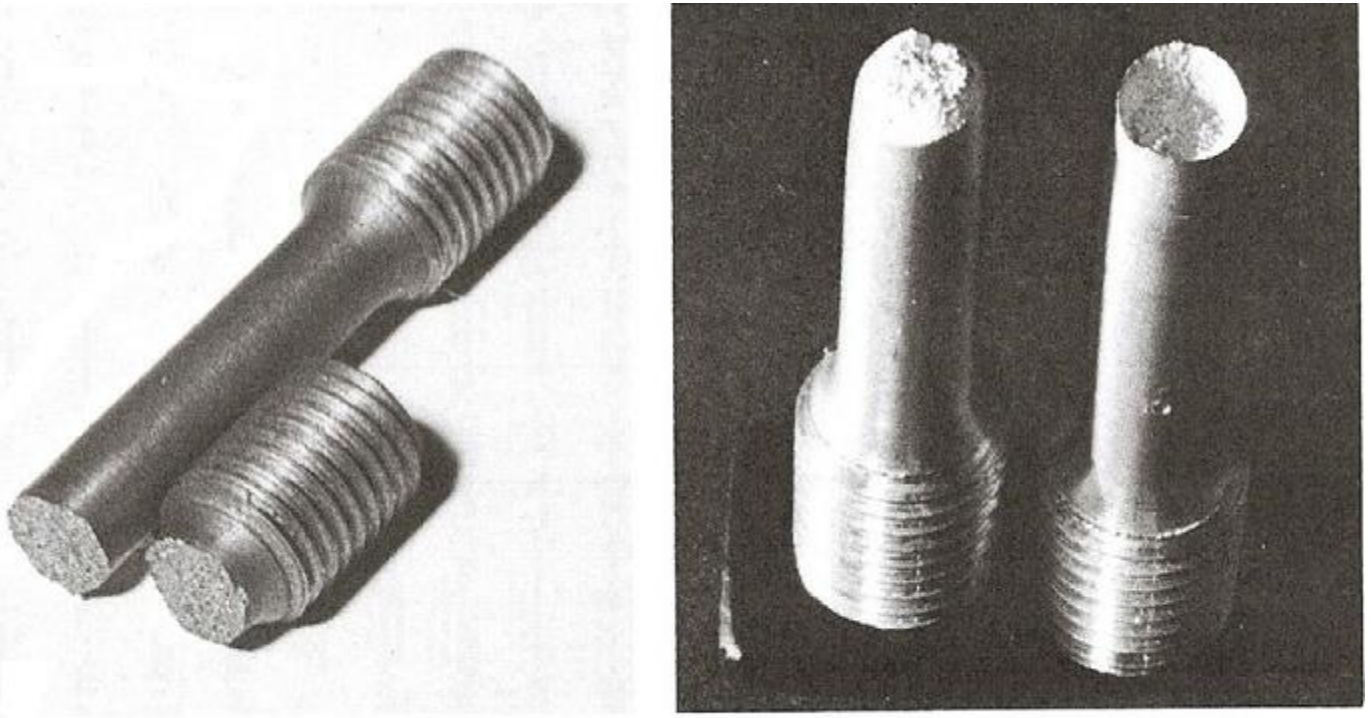




Stress-strain diagram for a typical brittle material.

Materials that exhibit very little inelastic deformation. In other words, materials that fail in tension at relatively low values of strain are considered brittle. Brittle materials include concrete, stone, cast iron, glass and plaster.

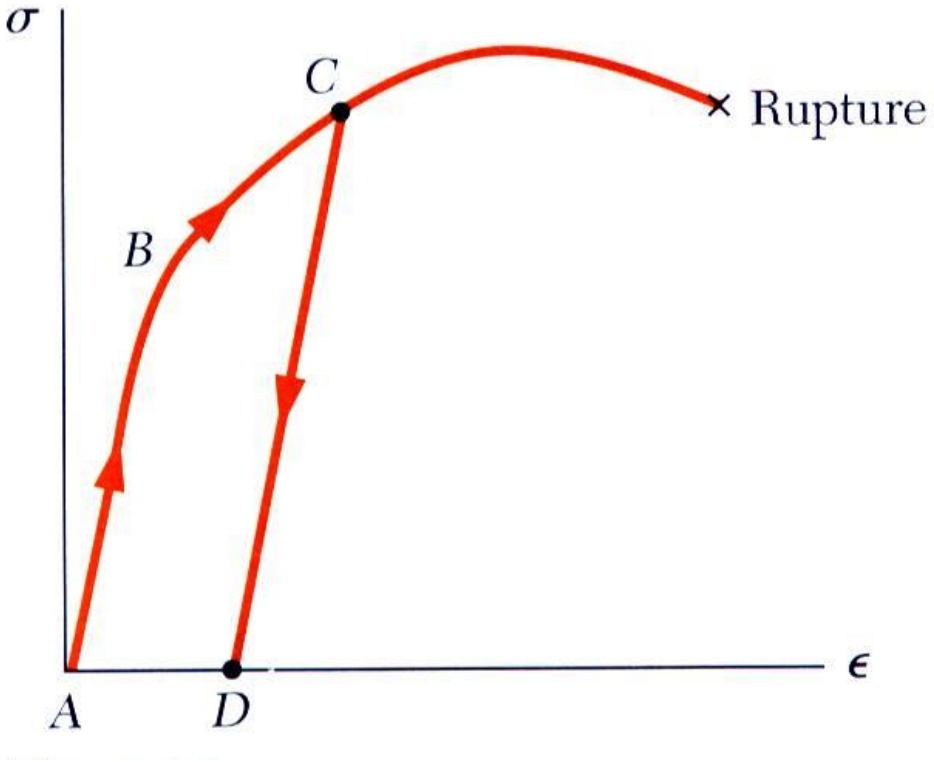
Failure modes



Typical Fracture Patterns of (a) Brittle and (b) Ductile Materials in Tension.

- **Brittle materials** fail due to tensile (*normal*) stresses and rupture occurs along a surface perpendicular to the load.
- **Ductile materials** usually fail on planes that correspond to the maximum shear stresses (*45°*). A cup and cone failure is typical for ductile materials with the sides of cup and cone inclined at approximately 45° to the specimen axis.

Elastic vs. Plastic Behavior



- If the strain disappears when the stress is removed, the material is said to behave *elastically*.
- The largest stress for which this occurs is called the *elastic limit*.
- When the strain does not return to zero after the stress is removed, the material is said to behave *plastically*.



Young observed that the σ-ε slope was a constant for a particular material, independent of the sample geometry.

STRESS ∝ STRAIN

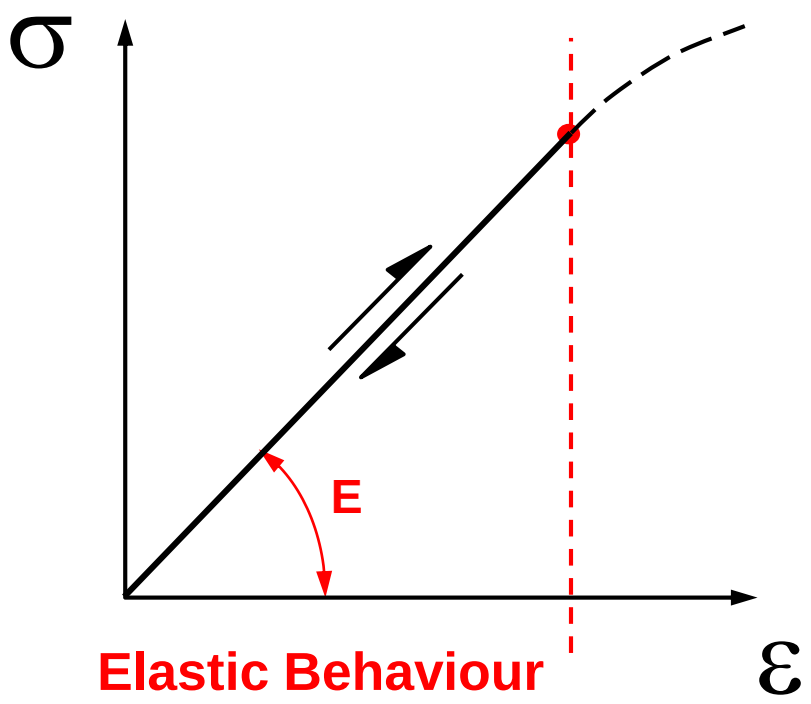
He defined an important material parameter, now known as the modulus of elasticity, or Young's modulus.

For uniaxial loading:

$$\Rightarrow \quad \sigma = \mathbf{E} \cdot \varepsilon$$

$$\text{or} \quad \mathbf{E} = \frac{\sigma}{\varepsilon}$$

Unit: Pa



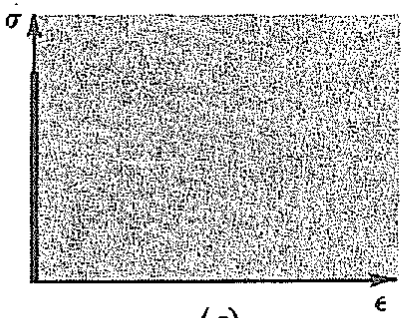
	Mild Stee	Aluminium	Concrete	Wood	Nylon	Rubber
Young's Mod, E GPa	210	70	18.5	12.5	2.8	0.004

Idealization of the stress-strain curve

- Depending on the problem at hand, there is a need for a variety of stress-strain relations
- It is impossible to describe the entire stress strain curve with a simple mathematical expression

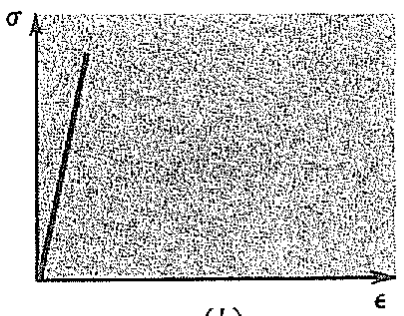
In any given problem, the behaviour of the material is represented by an *idealized stress-strain curve* which emphasizes those aspects of the behaviour which are most important in that particular porblem

Idealized models of behaviour



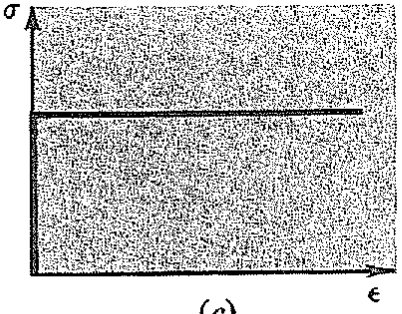
(*a*)

Rigid material.



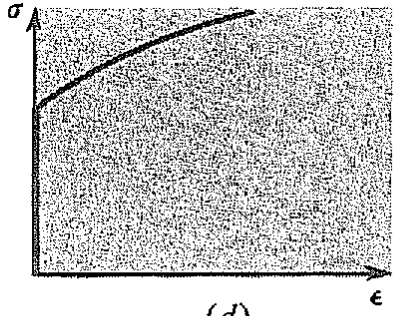
(*b*)

Linearly *elastic* material.



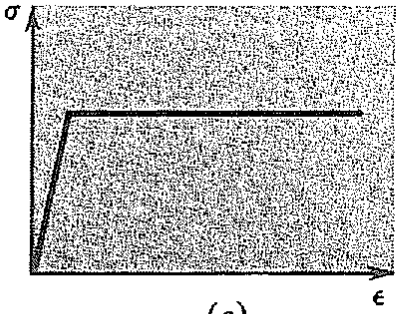
(*c*)

Perfectly plastic material (non-strain-hardening).



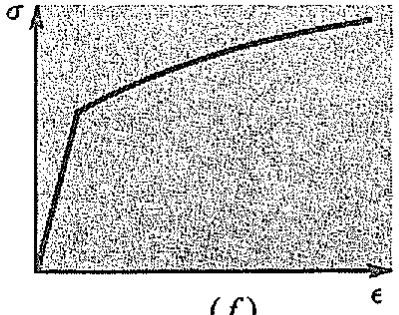
(*d*)

Rigid-plastic material (strain-hardening).



(*e*)

Elastic–perfectly plastic material (non-strain-hardening).



(*f*)

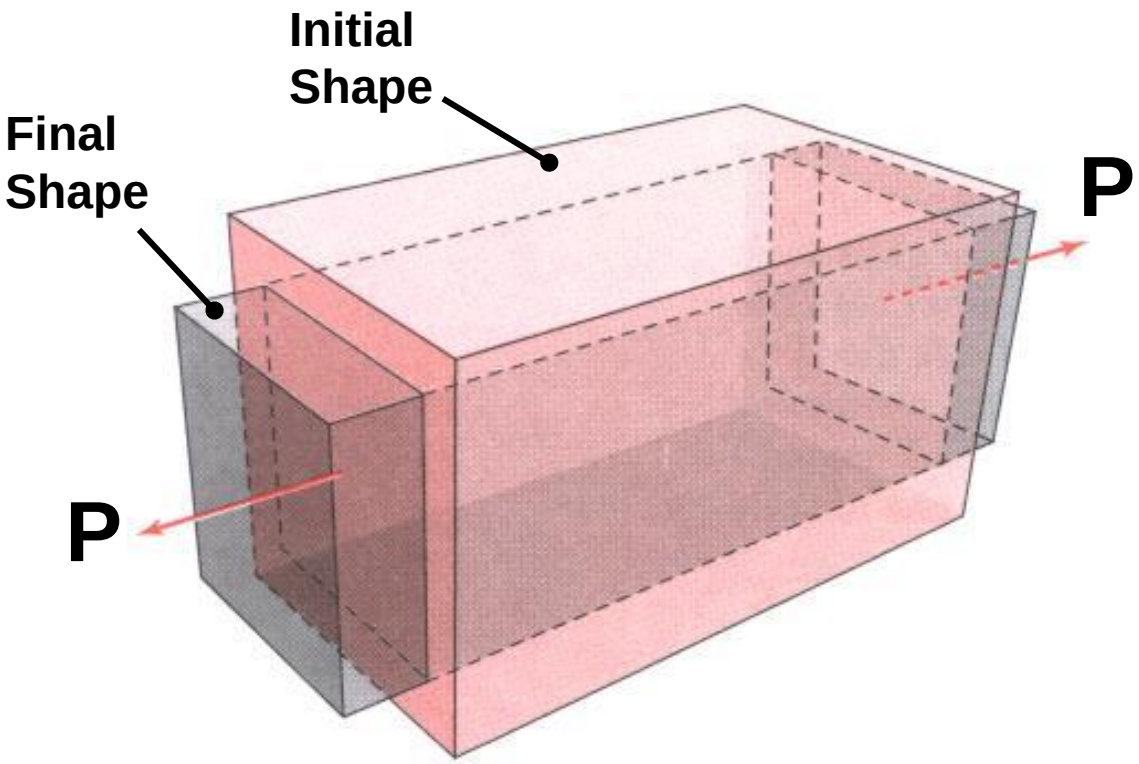
Elastic-plastic material (strain-hardening).

Material Properties (Cont.) (Simon Poisson, 1825)

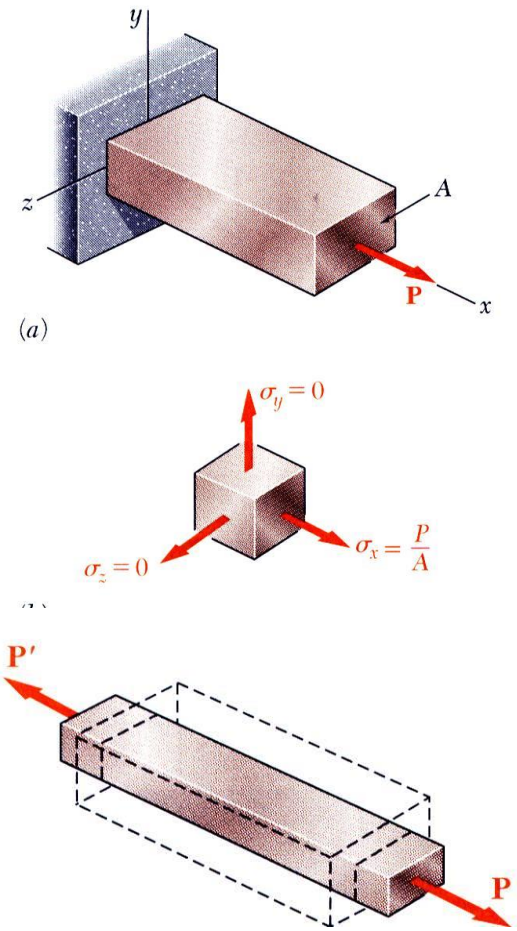
Poisson made important observations and theories about lateral deflections of materials.

When a bar is placed in tension, lateral contractions accompany the extension.





Poisson's Ratio



- For a slender bar subjected to axial loading: $\varepsilon_x = \dfrac{\sigma_x}{E} \quad \sigma_y = \sigma_z = 0$

- The elongation in the x-direction is accompanied by a contraction in the other directions. Assuming that the material is isotropic (no directional dependence),

$$\varepsilon_y = \varepsilon_z \neq 0$$

- Poisson's ratio is defined as

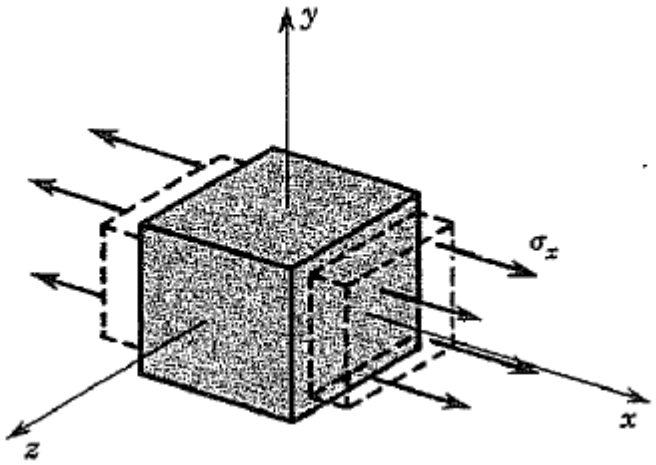
$$\nu = \left| \frac{\text{lateral strain}}{\text{axial strain}} \right| = -\frac{\varepsilon_y}{\varepsilon_x} = -\frac{\varepsilon_z}{\varepsilon_x}$$

ELASTIC STRESS STRAIN RELATIONS

Assumptions:

1. Material is linearly elastic
2. strain is linearly related to stress
3. deformations are small
4. material is isotropic

Uniaxial normal stress.

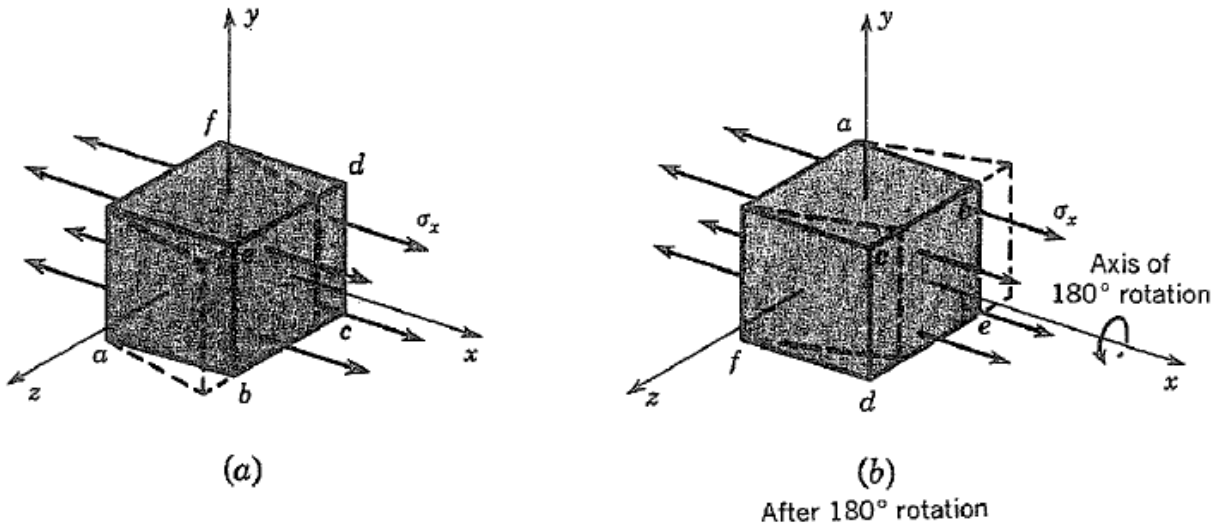


$$\epsilon_x = \frac{\sigma_x}{E}$$

$$\epsilon_y = \epsilon_z = -\nu\epsilon_x = -\nu\frac{\sigma_x}{E}$$

Possibility of shear strains from normal stress σ_x

Assumption of isotropy of material simplifies the relation
Suppose shear strain is present (Fig.a)



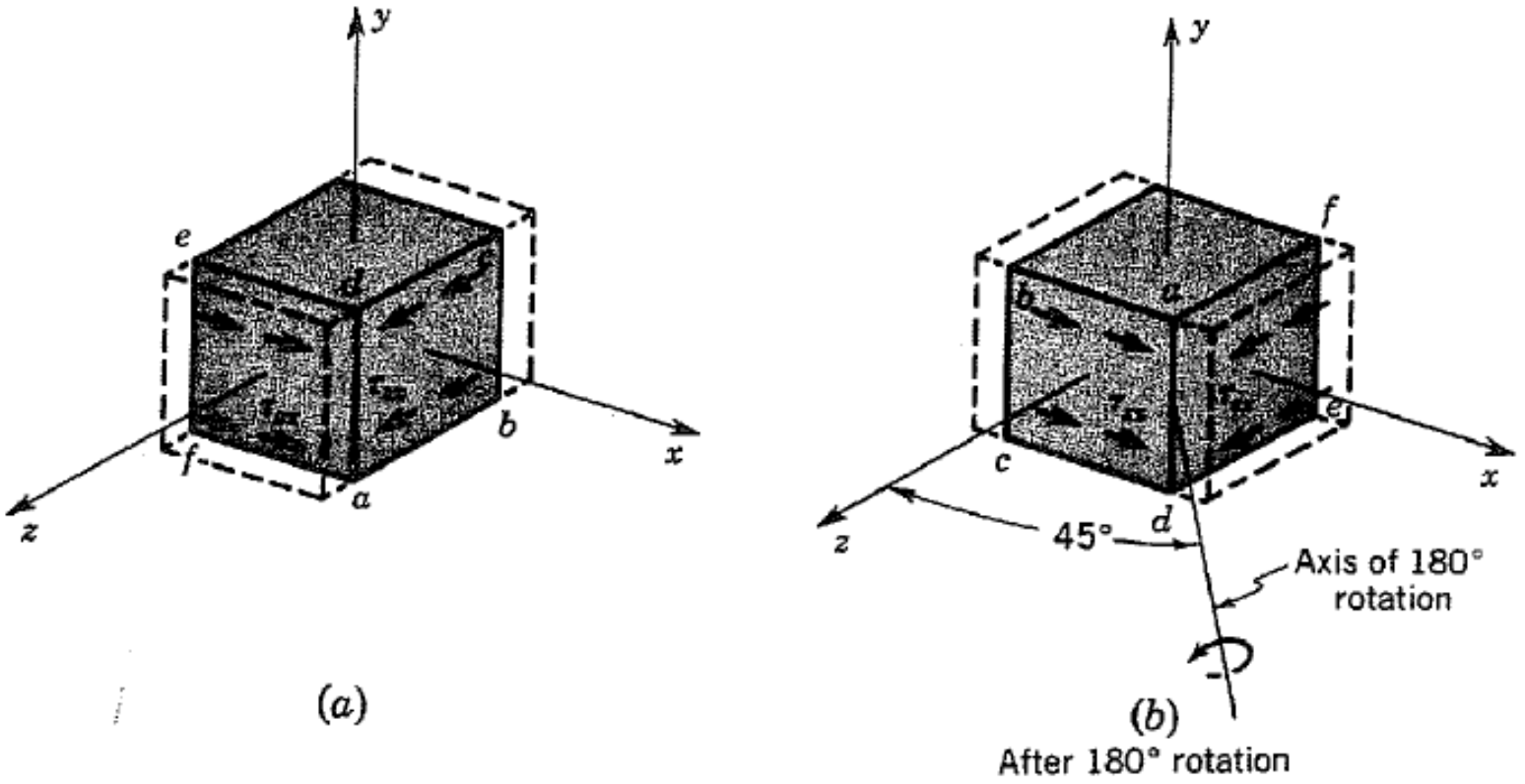
Normal stress produces only normal strain

Suppose normal stress σ_y is present

$$\epsilon_y = \frac{\sigma_y}{E}$$

$$\epsilon_x = \epsilon_z = -\nu\epsilon_y = -\nu\frac{\sigma_y}{E}$$

Possibility of ε_y due to shear stress τ_{zx}



Sign of shear stress changes but sign of hypothetical normal strain remains same, which violates the linearity of the material. Hence normal strain is absent due to shear stress alone

Shear stress gives rise to shear strain only

$$\gamma_{zx} = \frac{\tau_{zx}}{G} \qquad \gamma_{xy} = \frac{\tau_{xy}}{G} \qquad \gamma_{yz} = \frac{\tau_{yz}}{G}$$

G is called the *shear modulus*.

Generalized Hooke's Law

- For an element subjected to multi-axial loading, the normal strain components resulting from the stress components may be determined from the *principle of superposition*. This requires:

1. strain is linearly related to stress
2. deformations are small
3. material is isotropic

- With these restrictions:

$$\varepsilon_x = +\frac{\sigma_x}{E} - \frac{\nu\sigma_y}{E} - \frac{\nu\sigma_z}{E}$$

$$\varepsilon_y = -\frac{\nu\sigma_x}{E} + \frac{\sigma_y}{E} - \frac{\nu\sigma_z}{E}$$

$$\varepsilon_z = -\frac{\nu\sigma_x}{E} - \frac{\nu\sigma_y}{E} + \frac{\sigma_z}{E}$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

$$\gamma_{yz} = \frac{\tau_{yz}}{G}$$

$$\gamma_{zx} = \frac{\tau_{zx}}{G}$$

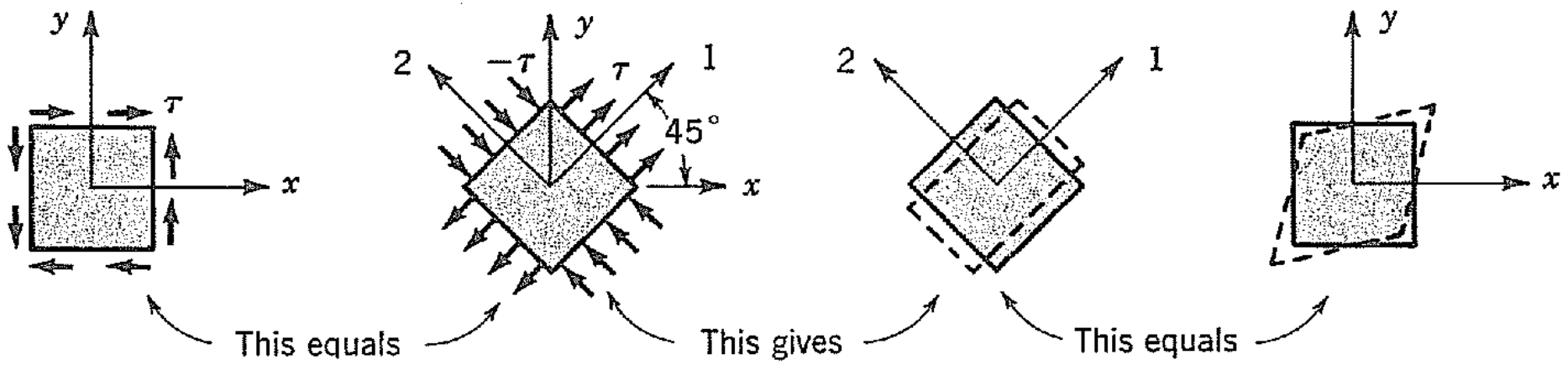
Relation Among E, ν, and G

Due to isotropy of material: *The principal axes of strain at a point of a stresses body coincide with the principal axes of stress at that point.*

Hence in determining the location of the principal axes corresponding to given state of stress, one may use either the Mohr's circle for stress or that for strain

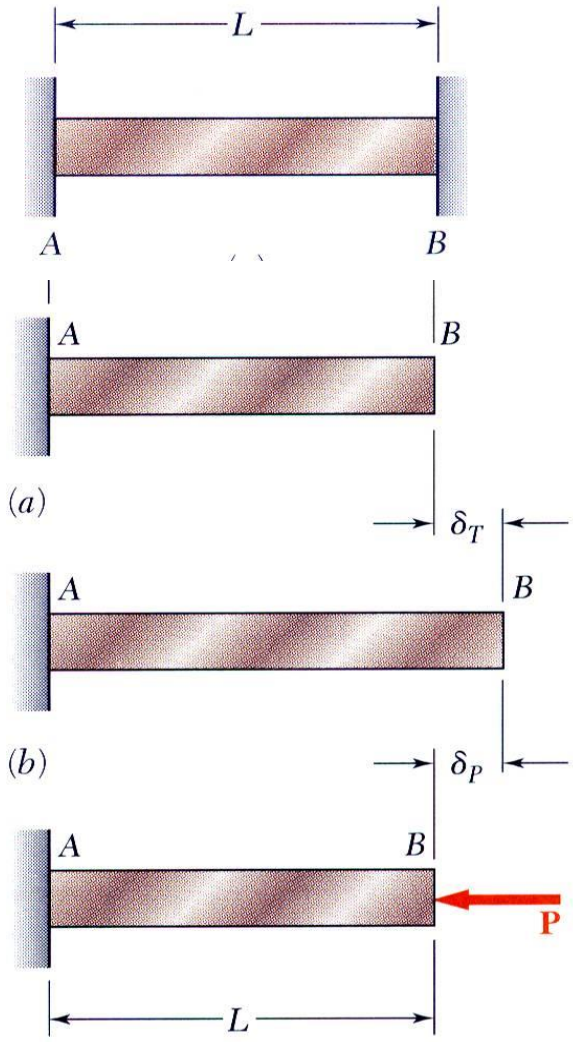
Isotropic consideration of material also can be used to find the relations between three elastic constants.

Relation Among E, ν, and G



$$\frac{E}{2G} = (1+\nu)$$

Thermal Strain



- A temperature change results in a change in length or *thermal strain*. There is no stress associated with the thermal strain unless the elongation is restrained by the supports.

$$\varepsilon_t = \alpha(T - T_0) = \alpha \Delta T$$

- Treat the additional support as redundant and apply the principle of superposition.

$$\delta_T = \alpha(\Delta T)L \qquad \delta_P = \frac{PL}{AE}$$

$\alpha =$ thermal expansion coef.

- The thermal deformation and the deformation from the redundant support must be compatible.

$$\delta = \delta_T + \delta_P = 0$$

$$\alpha(\Delta T)L + \frac{PL}{AE} = 0$$

$$\delta = \delta_T + \delta_P = 0$$
$$P = -AE\alpha(\Delta T)$$

$$\sigma = \frac{P}{A} = -E\alpha(\Delta T)$$

Complete equations of elasticity

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + X = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + Y = 0$$

$$\frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + Z = 0$$

Equilibrium

Geometric Compatibility

$$\epsilon_x = \frac{\partial u}{\partial x} \qquad \gamma_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

$$\epsilon_y = \frac{\partial v}{\partial y} \qquad \gamma_{yz} = \frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}$$

$$\epsilon_z = \frac{\partial w}{\partial z} \qquad \gamma_{zx} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}$$

Complete equations of elasticity

$$\epsilon_x = \frac{1}{E}[\sigma_x - \nu(\sigma_y + \sigma_z)] + \alpha(T - T_o)$$

$$\epsilon_y = \frac{1}{E}[\sigma_y - \nu(\sigma_z + \sigma_x)] + \alpha(T - T_o)$$

$$\epsilon_z = \frac{1}{E}[\sigma_z - \nu(\sigma_x + \sigma_y)] + \alpha(T - T_o)$$

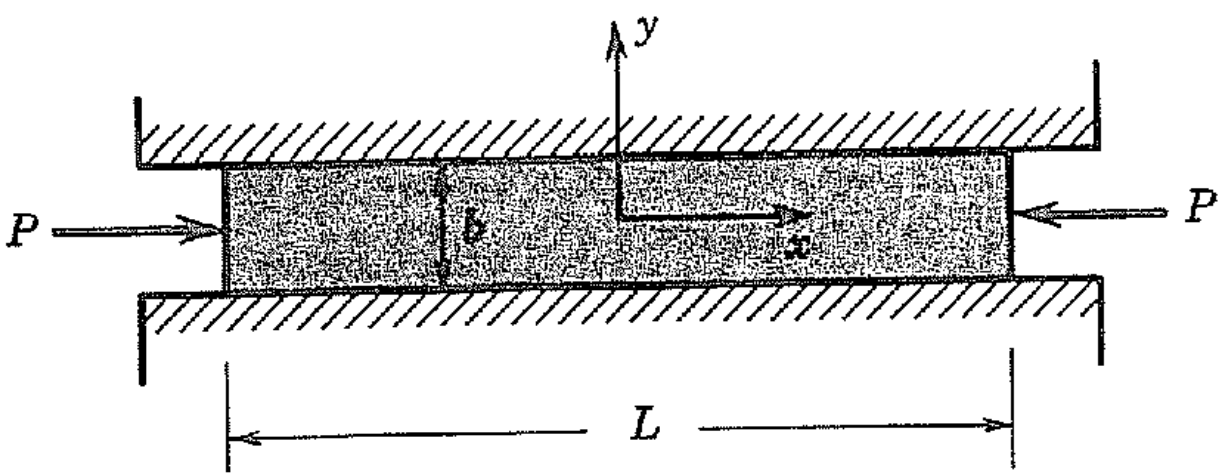
$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

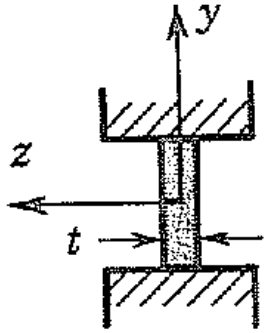
$$\gamma_{yz} = \frac{\tau_{yz}}{G}$$

$$\gamma_{zx} = \frac{\tau_{zx}}{G}$$

Stress strain temperature relations

Problems



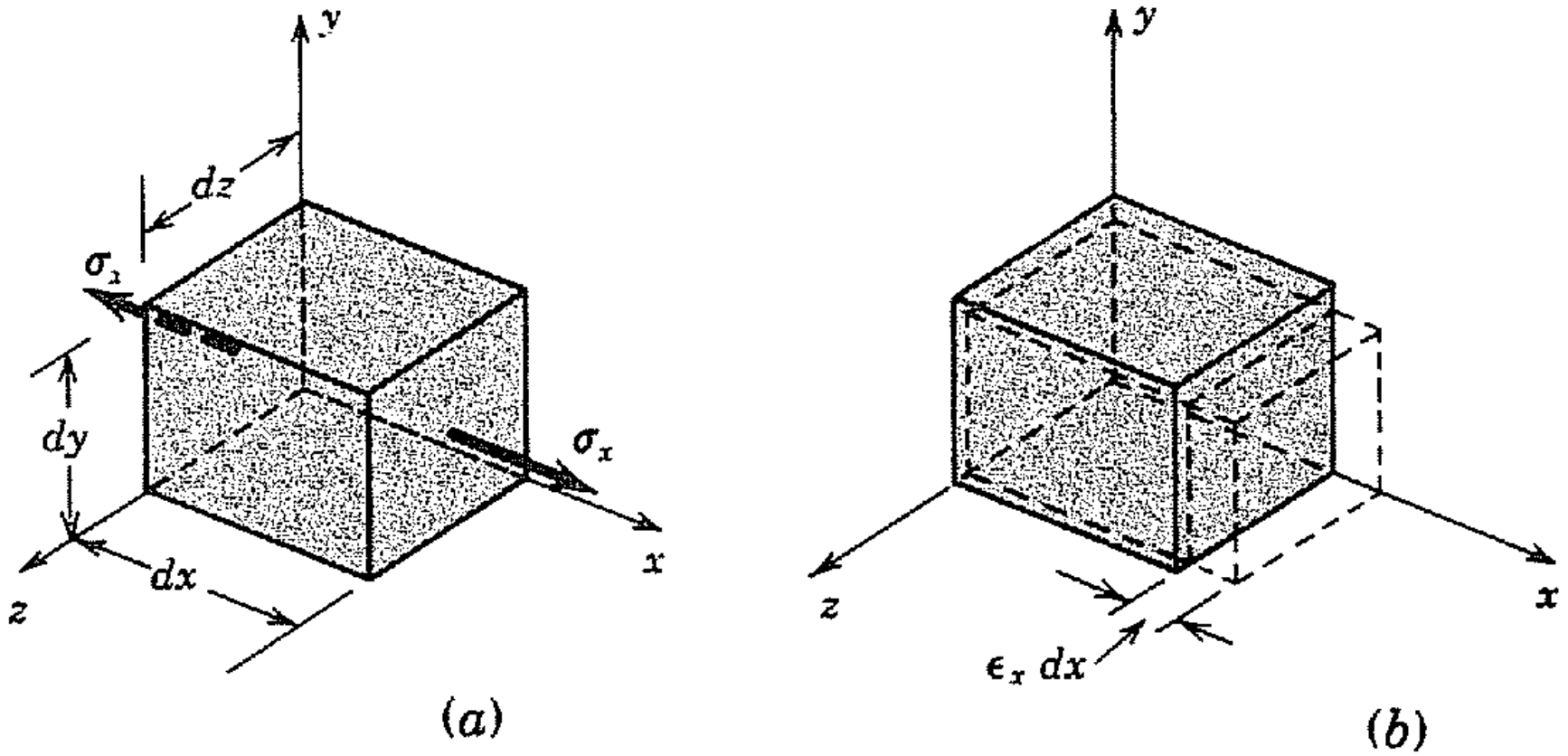


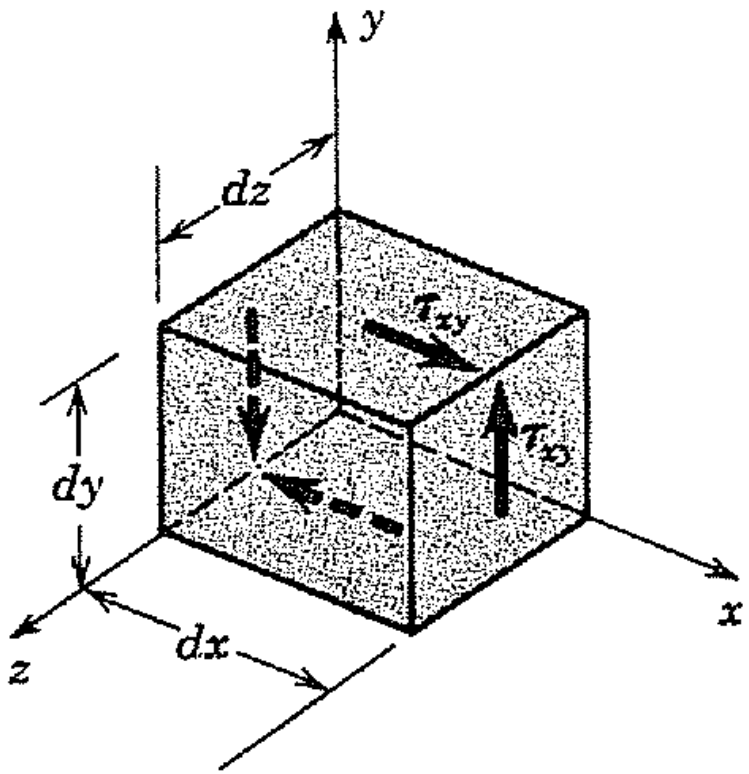
A long, thin plate of width b, thickness t, and length L is placed between two rigid walls a distance b apart and is acted on by an axial force P. *Find the deflection of the plate parallel to the force P.*

Assumptions:

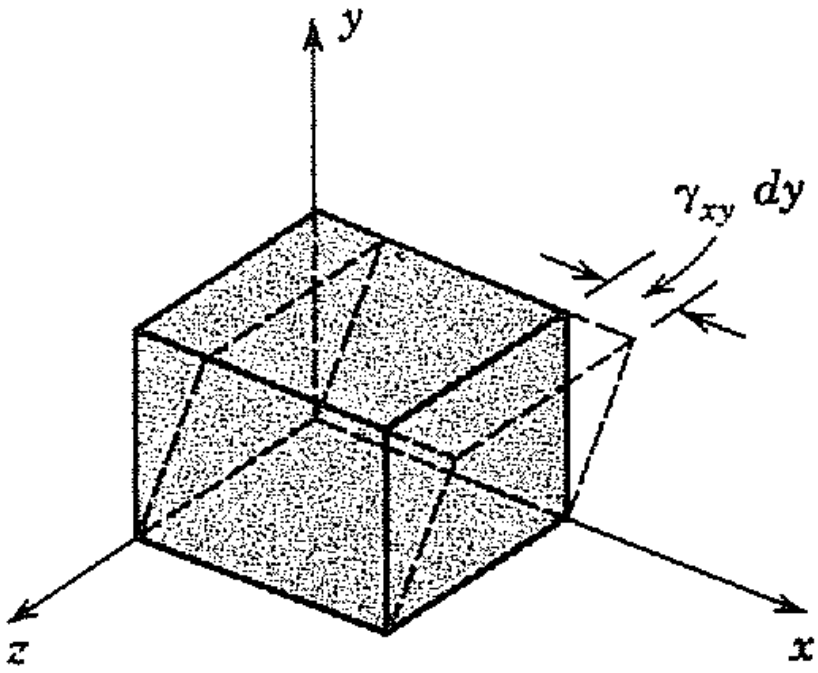
- No normal stress in z-direction.(plane stress condition in xy plane)
- $\varepsilon_y = 0$ (plane strain condition in xz plane)
- Negligible or no friction at the walls

Strain energy in an elastic body





(c)



(d)

Background and definitions

Yielding

- For ductile material under simple tension, stress no longer proportional to strain
- Plastic (irreversible) deformation (permanent molecular rearrangement) once a certain level of stress is reached
- Highly material dependent

Understanding yielding is important for designing a pressure vessel, rotating disc, crank shaft, ... that does not allow any irreversible strain, i.e. material must remain elastic

Fracture vs yield

Fracture

- Driven by normal stresses, acting to separate one atomic plane from another
- Broken atomic bonds are not allowed to reform in new positions

Yield

- Driven by shear stresses, sliding one plane along another
- Broken atomic bonds are allowed to reform in new positions

Yield criteria

- For material stretched uniaxially along $\mathbf{e}_1$ direction, yield occurs when :

$$\boxed{\sigma_{11} \geq \sigma_y}$$

with σ_y is the yield stress

- When does yield occurs in multiaxial stress states...??

Tresca Criterion

General multiaxial stress states

Maximum shear stress

Yielding starts when the maximum shear stress in the material τ_{max} equals the maximum shear stress at yielding in a simple tension test τ_y

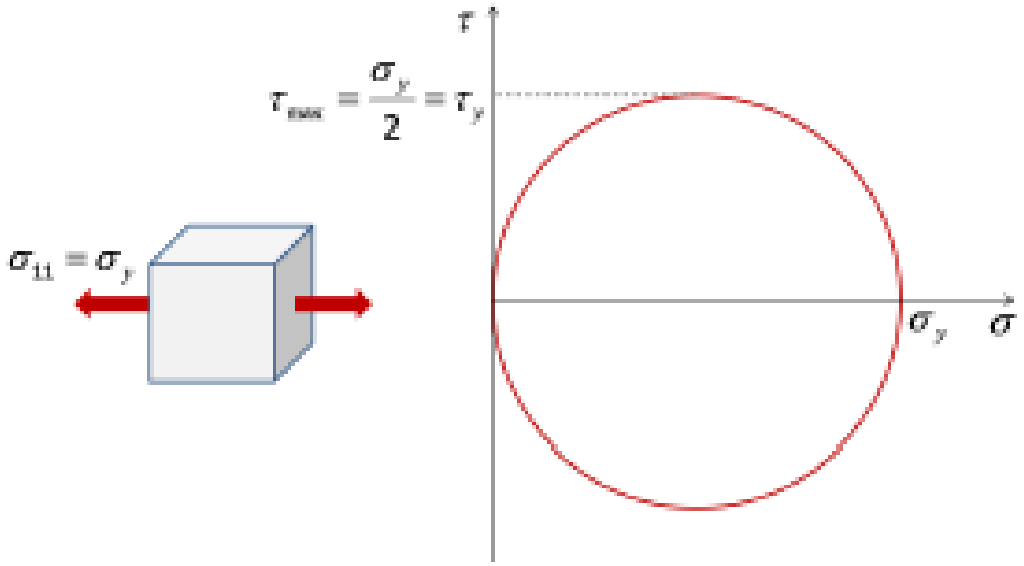
$$\boxed{\tau_{max} = \tau_y}$$

where : $\tau_{max} = \frac{\sigma_{max} - \sigma_{min}}{2}$

σ_{max} and σ_{min} are the maximum and minimum principal stresses respectively

General multiaxial stress states

Mohr's circle for simple tension test :



Thus, general form of Tresca Criterion is :

$$\sigma_{\max} - \sigma_{\min} = \sigma_y$$

Special case : Plane stress

Let σ_1, σ_2 and σ_3 be the principale stresses $(\sigma_3 = 0)$:

- When σ_1 and σ_2 are of opposite sign : $\tau_{\max} = \frac{|\sigma_1 - \sigma_2|}{2}$
  The yield condition is given by :

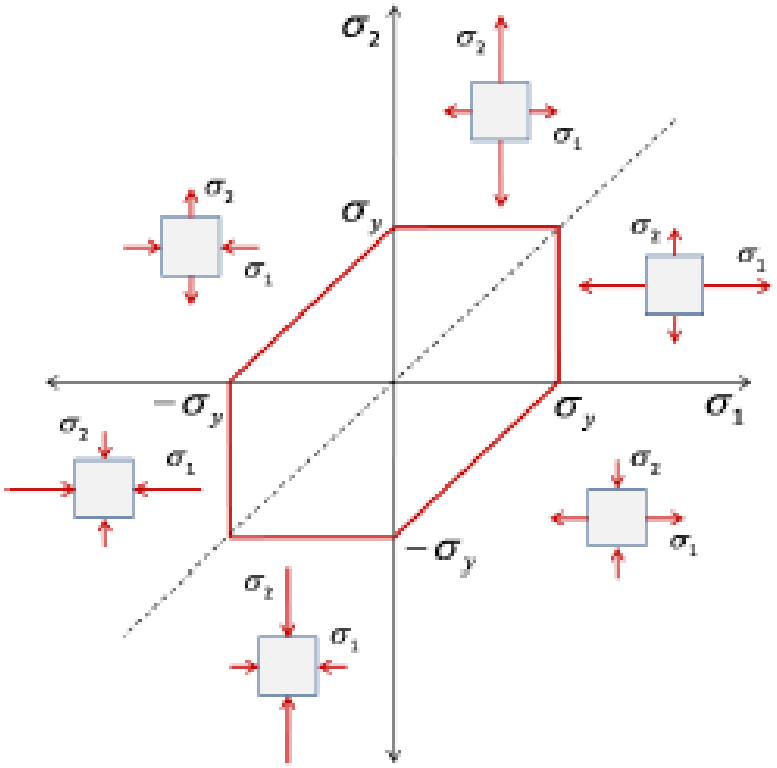
$$|\sigma_1 - \sigma_2| = \sigma_y \qquad \text{or} \qquad \frac{\sigma_1}{\sigma_y} - \frac{\sigma_2}{\sigma_y} = \pm 1$$

- When σ_1 and σ_2 carry the same sign :

$$\text{if} \quad |\sigma_1| > |\sigma_2|, \qquad \tau_{\max} = \frac{|\sigma_1 - \sigma_3|}{2} = \frac{|\sigma_1|}{2} \quad \text{and} \quad |\sigma_1| = \sigma_y$$

$$\text{if} \quad |\sigma_1| < |\sigma_2|, \qquad \tau_{\max} = \frac{|\sigma_2 - \sigma_3|}{2} = \frac{|\sigma_2|}{2} \quad \text{and} \quad |\sigma_2| = \sigma_y$$

Tresca yield surface for plane stress problems

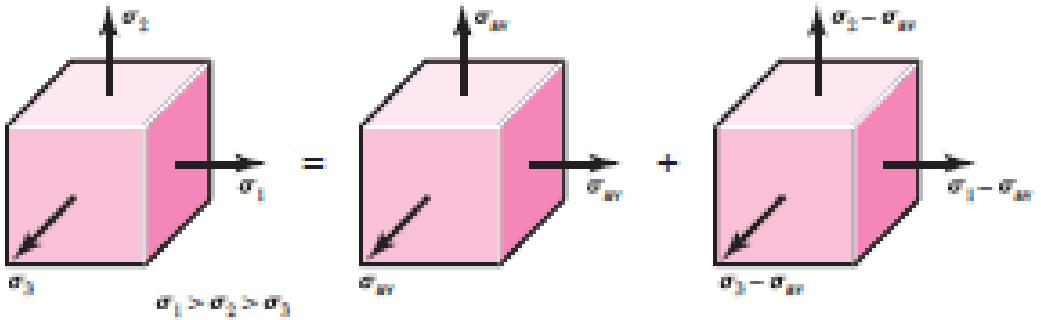


Von Mises Criterion

The *distortion-energy theory* predicts that *yielding occurs when the distortion strain energy per unit volume reaches or exceeds the distortion strain energy per unit volume for yield in simple tension or compression of the same material.*

$$\boxed{W_{d,\max} = W_{d,y}}$$

The distortion-energy (DE) theory originated from the observation that ductile materials stressed hydrostatically (equal principal stresses) exhibited yield strengths greatly in excess of the values given by the simple tension test. Therefore it was postulated that yielding was not a simple tensile or compressive phenomenon at all, but, rather, that it was related somehow to the angular distortion of the stressed element.



(*a*) Triaxial stresses (*b*) Hydrostatic component (*c*) Distortional component

$$\sigma_{av} = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}$$

The strain energy per unit volume for simple tension is $u = \frac{1}{2}\epsilon\sigma$. For the element of Fig. a the strain energy per unit volume is $u = \frac{1}{2}[\epsilon_1\sigma_1 + \epsilon_2\sigma_2 + \epsilon_3\sigma_3]$. Substituting for the principal strains gives

$$u = \frac{1}{2E}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)\right]$$

The strain energy for producing only volume change u_v can be obtained by substituting σ_{av} for σ_1, σ_2, and σ_3 . The result is

$$u_v = \frac{3\sigma_{av}^2}{2E}(1 - 2\nu)$$

Substituting σav

$$u_v = \frac{1 - 2\nu}{6E}\left(\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + 2\sigma_1\sigma_2 + 2\sigma_2\sigma_3 + 2\sigma_3\sigma_1\right)$$

Distortion energy is given by

$$u_d = u - u_v = \frac{1 + \nu}{3E}\left[\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}\right]$$ -------1

For the simple tensile test, at yield, $\sigma_1 = S_y$ and $\sigma_2 = \sigma_3 = 0$, the distortion energy is

$$u_d = \frac{1+\nu}{3E}\ \sigma y^2 \qquad \text{-------2}$$

1=2 gives the yield criterian

$$\left[\frac{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2}{2}\right]^{1/2} \geq \sigma y$$

Left hand side : the Von Mises stress σ_{vm} σy

LHS: VON MISSES STRESS $\sigma_{vm} \geq \sigma_y$

Special case : Plane stress

Let σ_1, σ_2 and σ_3 be the principale stresses $(\sigma_3 = 0)$:

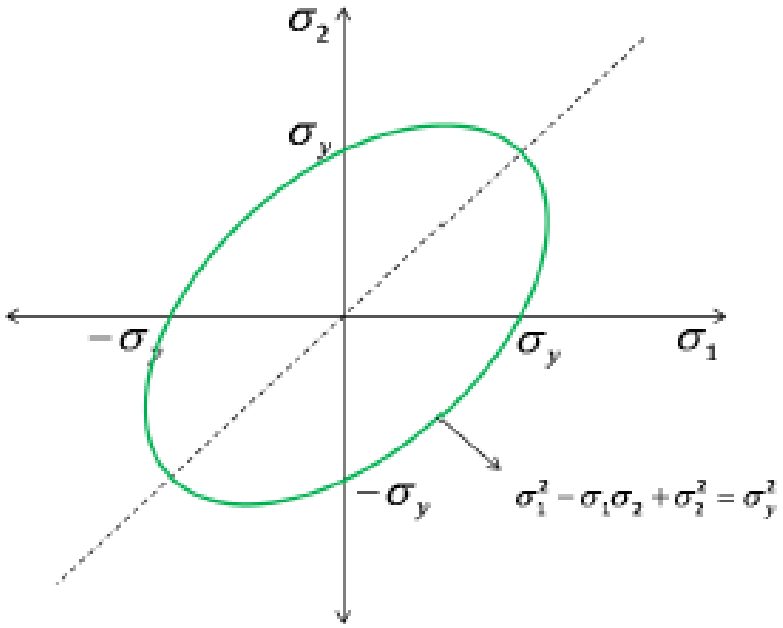
$$\begin{aligned} \sigma_{\text{vm}} &= \frac{1}{\sqrt{2}} \left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - 0)^2 + (0 - \sigma_1)^2 \right]^{1/2} \\ &= \sqrt{\sigma_1^2 - \sigma_1 \sigma_2 + \sigma_2^2} \end{aligned}$$

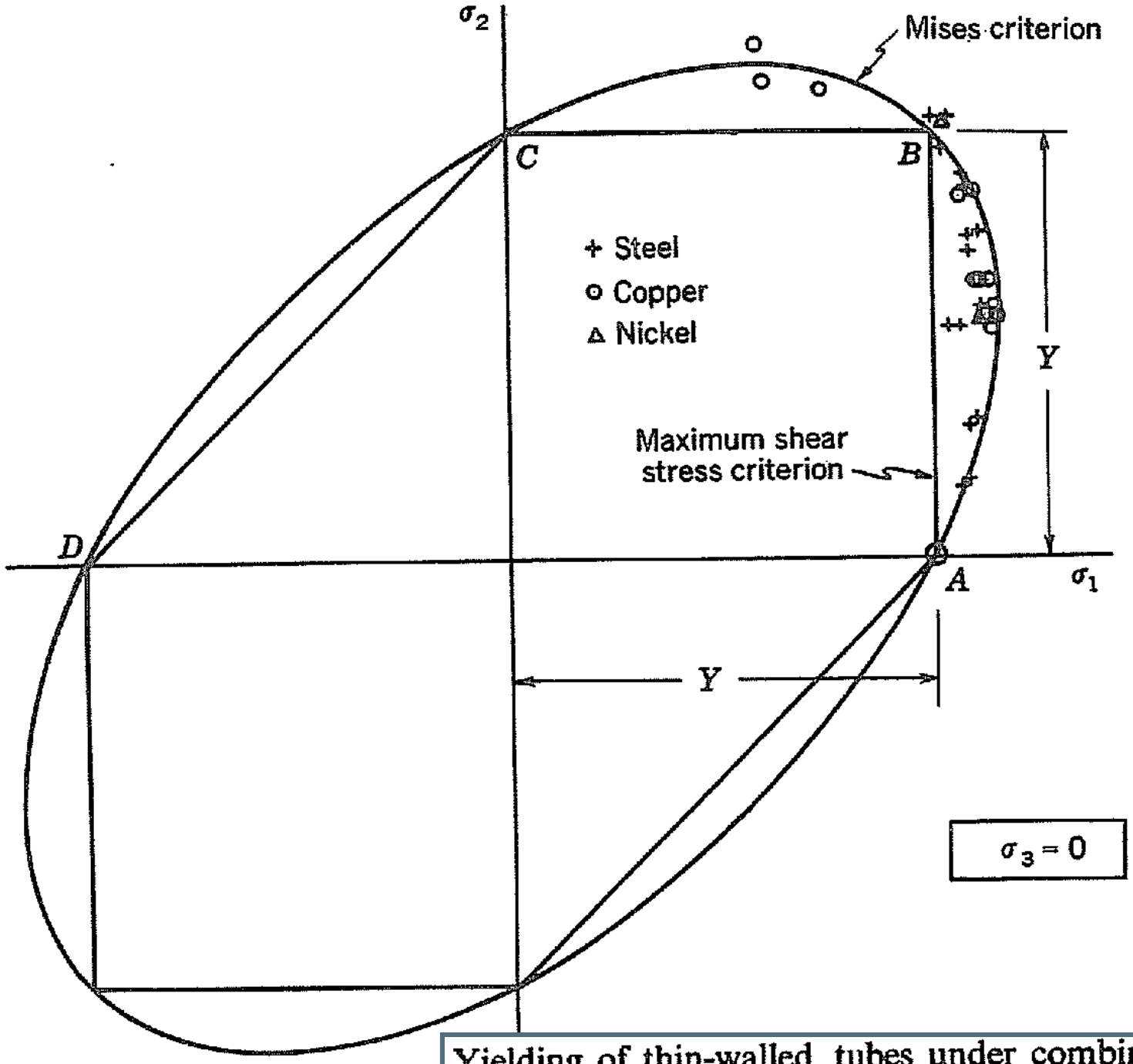
Von Mises yield criterion becomes :

$$\boxed{\sigma_1^2 - \sigma_1 \sigma_2 + \sigma_2^2 = \sigma_y^2}$$

In $\sigma_1 - \sigma_2$ plane, this equation represents an ellipse

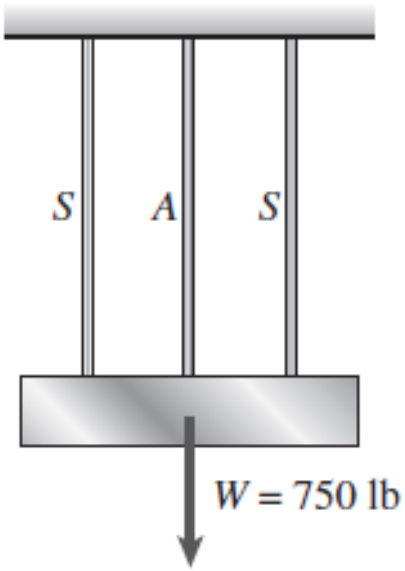
Von Misses yield surface for plane stress problems





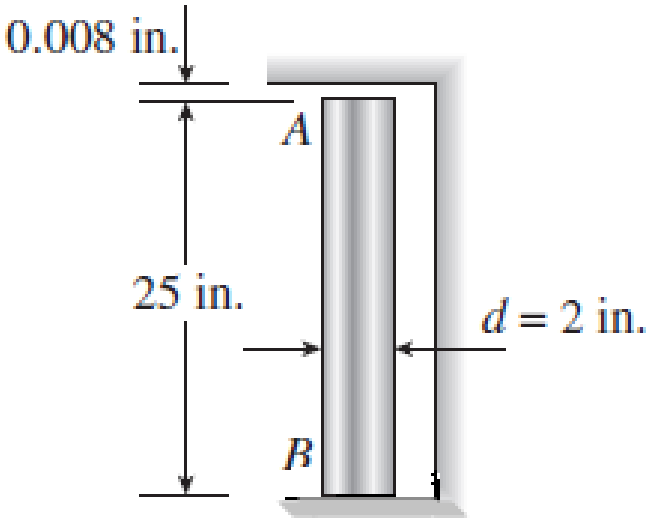
Yielding of thin-walled tubes under combined stress.

Problem



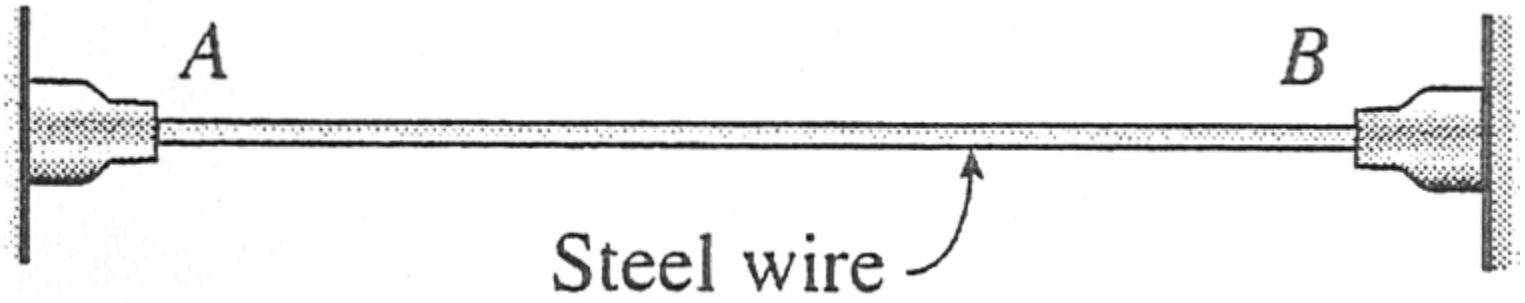
- A rigid bar of weight W 750 lb hangs from three equally spaced wires, two of steel and one of aluminum. The diameter of the wires is 1/8 in . Before they were loaded, all three wires had the same length.
- What temperature increase T in all three wires will result in the entire load being carried by the steel wires? (Assume $E_s = 30 \times 10^6$ psi, $\alpha_s = 6.5 \times 10^{-6}/°F$, and $\alpha_a = 12 \times 10^{-6}/°F$.)

Problem



- A copper bar AB of length 25 in. is placed in position at room temperature with a gap of 0.008 in. between end A and a rigid restraint (see figure).
- Calculate the axial compressive stress σ_c in the bar if the temperature rises 50°F. (For copper, use $\alpha = 9.6 \times 10^{-6}/°F$ and $E = 16 \times 10^6$ psi.)

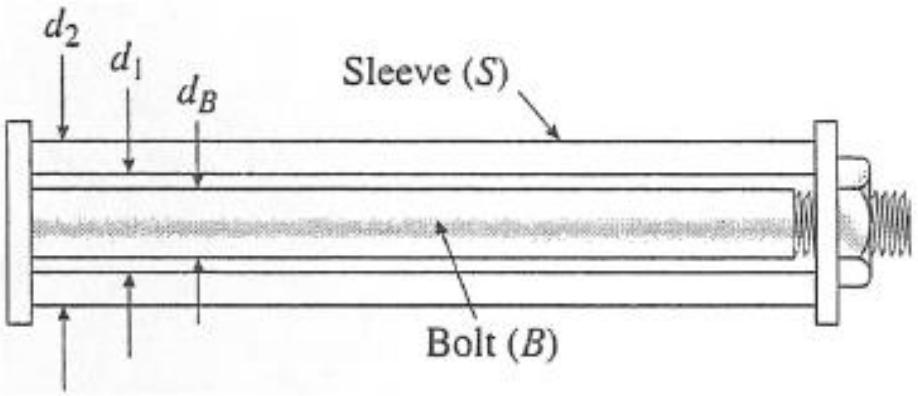
Problem

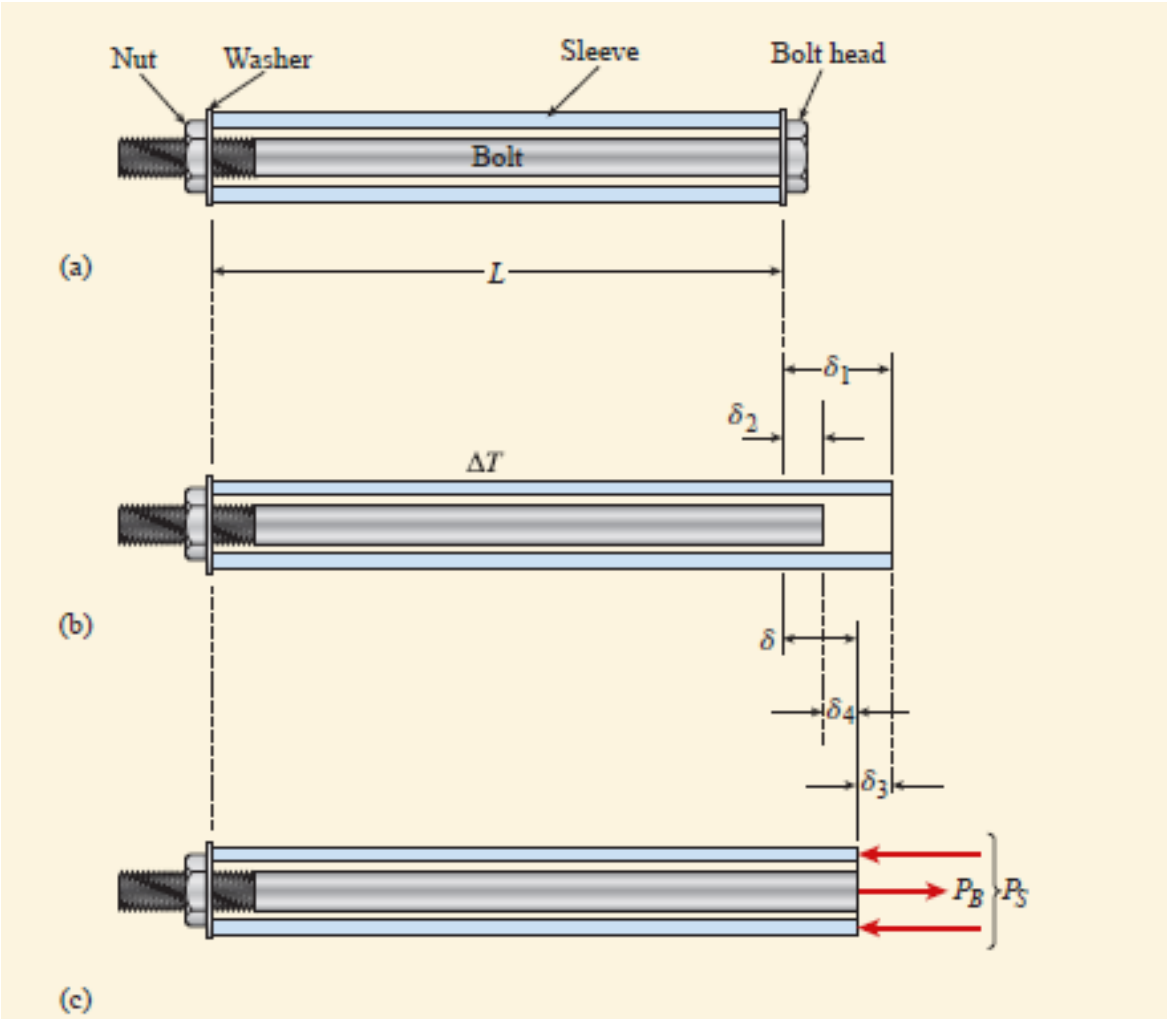


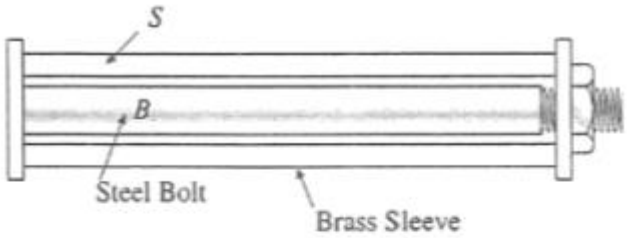
- A steel wire AB is stretched between rigid supports. The initial pre-stress in the wire is 42 MPa when the temperature is 20°C.
- (a) What is the stress in the wire when the temperature drops to 0°C?
- (b) At what temperature T will the stress in the wire become zero?
- (Assume $\alpha = 14 \times 10^{-6}/°C$ and E=200 GPa.)

Problem A brass sleeve S is fitted over a steel bolt B (see figure), and the nut is tightened until it is just snug. The bolt has a diameter $d_B = 25$ mm, and the sleeve has inside and outside diameters $d_1 = 26$ mm and $d_2 = 36$ mm, respectively.

Calculate the temperature rise ΔT that is required to produce a compressive stress of 25 MPa in the sleeve. (Use material properties as follows: for the sleeve, $\alpha_S = 21 \times 10^{-6}/°C$ and $E_S = 100$ GPa; for the bolt, $\alpha_B = 10 \times 10^{-6}/°C$ and $E_B = 200$ GPa.)




Nut
Washer
Sleeve
Bolt head
Bolt
(a)
L
δ_1
δ_2
ΔT
(b)
δ
δ_4
δ_3
P_B
P_S
(c)



Subscript S means "sleeve".

Subscript B means "bolt".

σ_S = compressive force in sleeve

$$\sigma_S = \frac{(\alpha_S - \alpha_B)(\Delta T)E_S E_B A_B}{E_S A_S + E_B A_B} \text{ (Compression)}$$

SOLVE FOR ΔT:

$$\Delta T = \frac{\sigma_S(E_S A_S + E_B A_B)}{(\alpha_S - \alpha_B)E_S E_B A_B}$$

or

$$\Delta T = \frac{\sigma_S}{E_S(\alpha_S - \alpha_B)}\left(1 + \frac{E_S A_S}{E_B A_B}\right) \leftarrow$$

SUBSTITUTE NUMERICAL VALUES:

$\sigma_S = 25$ MPa

$d_2 = 36$ mm $\quad d_1 = 26$ mm $\quad d_B = 25$ mm

$E_S = 100$ GPa $\quad E_B = 200$ GPa

$\alpha_S = 21 \times 10^{-6}/°\text{C}$ $\quad \alpha_B = 10 \times 10^{-6}/°\text{C}$

$$A_S = \frac{\pi}{4}(d_2^2 - d_1^2) = \frac{\pi}{4}(620 \text{ mm}^2)$$

$$A_B = \frac{\pi}{4}(d_B)^2 = \frac{\pi}{4}(625 \text{ mm}^2)\ 1 + \frac{E_S A_S}{E_B A_B} = 1.4$$

$$\Delta T = \frac{25 \text{ MPa } (1.496)}{(100 \text{ GPa})(11 \times 10^{-6}/°\text{C})}$$

$\Delta T = 34°\text{C}$ $\leftarrow$

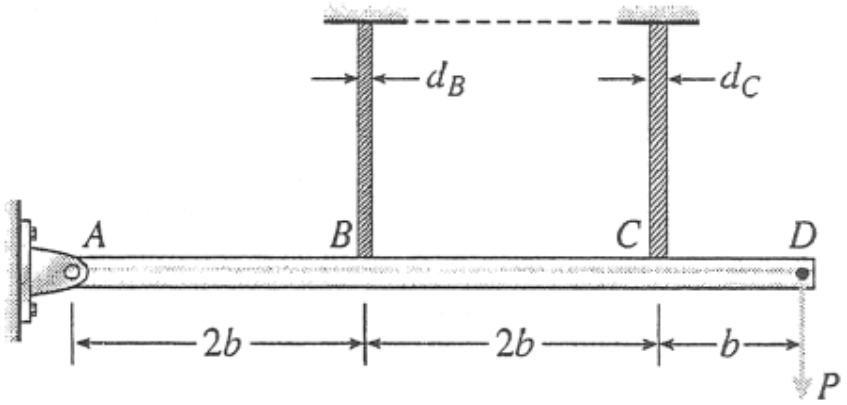
(Increase in temperature)

Problem A rigid bar $ABCD$ is pinned at end A and supported by two cables at points B and C (see figure). The cable at B has nominal diameter $d_B = 12$ mm and the cable at C has nominal diameter $d_C = 20$ mm. A load P acts at end D of the bar.

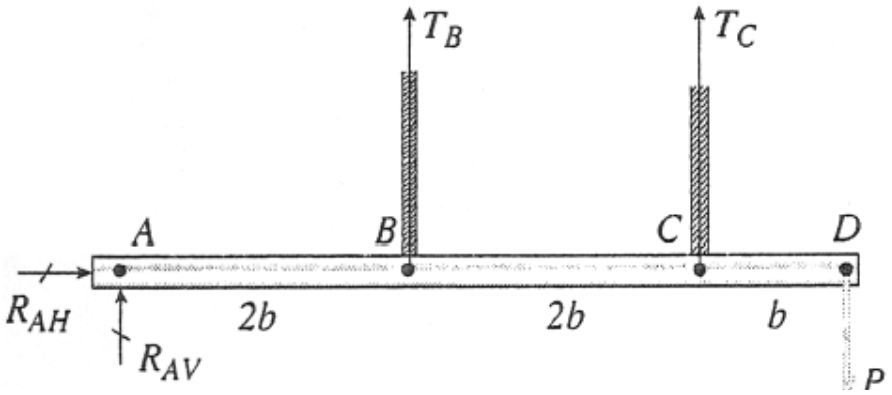
What is the allowable load P if the temperature rises by 60°C and each cable is required to have a factor of safety of at least 5 against its ultimate load?

(*Note:* The cables have effective modulus of elasticity $E = 140$ GPa and coefficient of thermal expansion $\alpha = 12 \times 10^{-6}$/°C.)

$(T_B)_{ULT} = 102{,}000$ N $(T_C)_{ULT} = 231{,}00($



FREE-BODY DIAGRAM OF BAR *ABCD*

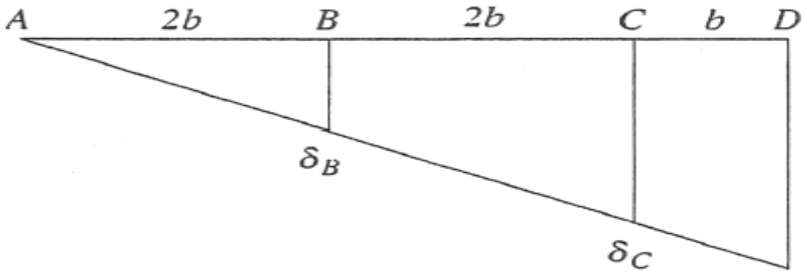


EQUATION OF EQUILIBRIUM

$$\Sigma M_A = 0 \quad \circlearrowleft \quad T_B(2b) + T_C(4b) - P(5b) = 0$$

or $2T_B + 4T_C = 5P$ (Eq. 1)

DISPLACEMENT DIAGRAM



COMPATIBILITY:

$$\delta_C = 2\delta_B \qquad \text{(Eq. 2)}$$

FORCE-DISPLACEMENT AND TEMPERATURE-DISPLACEMENT RELATIONS

$$\delta_B = \frac{T_B L}{EA_B} + \alpha(\Delta T)L \qquad \text{(Eq. 3)}$$

$$\delta_C = \frac{T_C L}{EA_C} + \alpha(\Delta T)L \qquad \text{(Eq. 4)}$$

SUBSTITUTE EQS. (3) AND (4) INTO EQ. (2):

$$\frac{T_C L}{E A_C} + \alpha(\Delta T)L = \frac{2T_B L}{E A_B} + 2\alpha(\Delta T)L$$

or

$$2T_B A_C - T_C A_B = -E\alpha(\Delta T)A_B A_C \qquad \text{(Eq. 5)}$$

SUBSTITUTE NUMERICAL VALUES INTO EQ. (5):

$$T_B(346) - T_C(76.7) = -1{,}338{,}000 \qquad \text{(Eq. 6)}$$

in which T_B and T_C have units of newtons.

SOLVE SIMULTANEOUSLY EQS. (1) AND (6):

$$T_B = 0.2494\,P - 3{,}480 \qquad \text{(Eq. 7)}$$

$$T_C = 1.1253\,P + 1{,}740 \qquad \text{(Eq. 8)}$$

in which P has units of newtons.

SOLVE EQS. (7) AND (8) FOR THE LOAD P:

$$P_B = 4.0096\,T_B + 13{,}953 \qquad \text{(Eq. 9)}$$

$$P_C = 0.8887\,T_C - 1{,}546 \qquad \text{(Eq. 10)}$$

$(T_B)_{\text{ULT}} = 102{,}000 \text{ N} \quad (T_C)_{\text{ULT}} = 231{,}000 \text{ N}$

Factor of safety $= 5$

$(T_B)_{\text{allow}} = 20{,}400 \text{ N} \quad (T_C)_{\text{allow}} = 46{,}200 \text{ N}$

From Eq. (9): $P_B = (4.0096)(20{,}400 \text{ N}) + 13{,}953 \text{ N}$
$= 95{,}700 \text{ N}$

From Eq. (10): $P_C = (0.8887)(46{,}200 \text{ N}) - 1546 \text{ N}$
$= 39{,}500 \text{ N}$

Cable C governs.

$P_{\text{allow}} = 39.5 \text{ kN} \quad \leftarrow$

- The load on a bolt subjected to an axial pull of 15 kN together with a shear of 7.5 kN. Determine the diameter of the bolt according to
- Tresca's criterion

- Mises yield criterion

- In uniaxial tension, material yields at the stress of 95 N/mm^2 and poisson's ratio is 0.3.